INCH-POUND

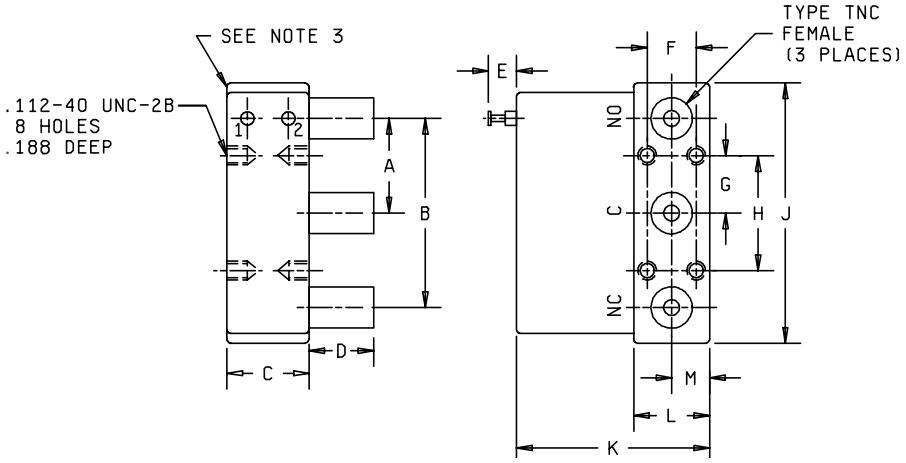
MIL-DTL-3928/20D
15 August 2001
SUPERSEDING
MIL-S-3928/20C
15 July 1988

DETAIL SPECIFICATION SHEET

SWITCHES, RADIO-FREQUENCY
TRANSMISSION LINE (COAXIAL) (ELECTRICALLY OPERATED)
CLASS 4, 1P2T

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Requirements for acquiring the switch described herein shall consist of this specification and MIL-DTL-3928.



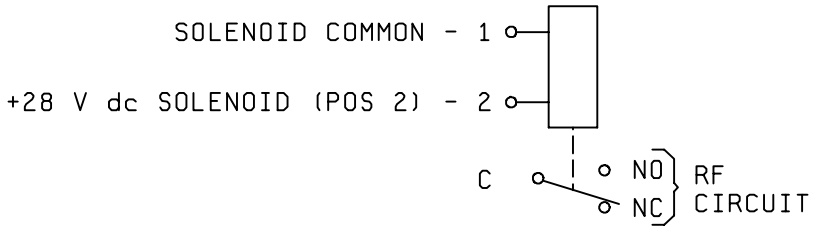
Dimensions									
Letter	Inches		Millimeters		Letter	Inches		Millimeters	
	Max	Min	Max	Min		Max	Min	Max	Min
A	.885	.865	22.48	21.97	G	.526	.506	13.36	12.85
B	1.78	1.72	45.2	43.7	H	1.042	1.022	26.47	25.96
C	.78	.72	19.8	18.3	J	2.41	2.35	61.2	59.7
D	.65	.59	16.5	15.0	K	1.72	1.66	43.7	42.2
E	.28	.22	7.1	5.6	L	.78	.72	19.8	18.3
F	.572	.552	14.53	14.02	M	.385	.365	9.78	9.27

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.
4. Round corners of case may be squared.

FIGURE 1. PIN M3928/20-02.

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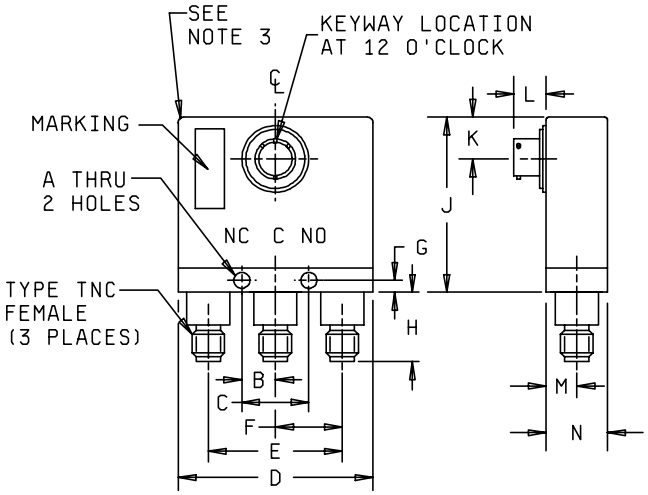
Position 1. (Deenergized or failsafe position),
C connects to NC connector.
NO is open circuit.
Position 2. (Energized position).
C connects to NO connector.
NC is open circuit.

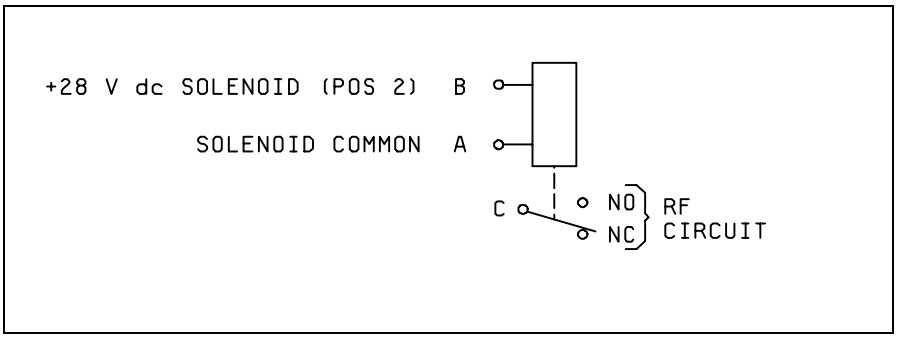
SCHEMATIC (SHOWN IN POSITION 1)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.
4. Round corners of case may be squared.

FIGURE 1. PIN M3928/20-02 - Continued.

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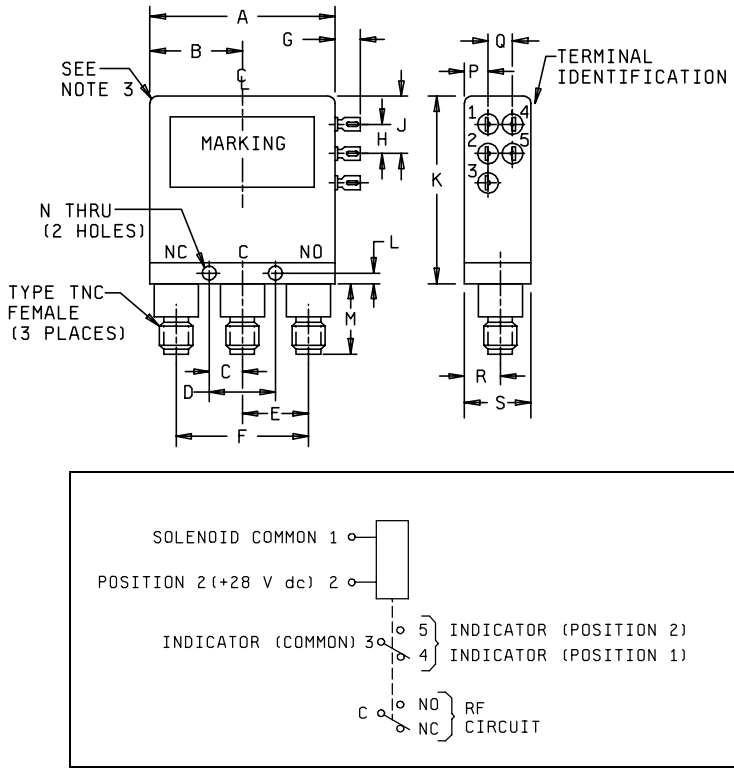
Position 1. (Deenergized or failsafe position), C connects to NC connector. NO is open circuit.
Position 2. (Energized position). C connects to NO connector. NC is open circuit.
SCHEMATIC (SHOWN IN POSITION 1)

Dimensions									
Letter	Inches		Millimeters		Letter	Inches		Millimeters	
	Max	Min	Max	Min		Max	Min	Max	Min
A	.213 dia	.193 dia	5.41 dia	4.90 dia	H	.56	---	14.2	---
B	.452	.432	11.48	10.97	J	2.07	---	52.6	---
C	.895	.875	22.73	22.22	K	.43	.37	10.9	9.4
D	2.65	---	67.3	---	L	.65	---	16.5	---
E	1.780	1.760	45.21	44.70	M	.541	.521	13.74	13.23
F	.895	.875	22.73	22.23	N	1.09	---	27.7	---
G	.210	.190	5.33	4.83					

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.
4. Round corners of case may be squared.

FIGURE 2. PIN M3928/20-03.

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Position 1. (Deenergized or failsafe position). C connects to NC connector. NO is open circuit.
Position 2. (Energized position). C connects to NO connector. NC is open circuit.

SCHEMATIC (SHOWN IN POSITION 1)

Letter	Inches		Millimeters		Letter	Inches		Millimeters	
	Max	Min	Max	Min		Max	Min	Max	Min
A	1.83	1.77	46.5	45.0	K	1.80	---	45.7	---
B	.93	.87	23.6	22.1	L	.12	.06	3.0	1.5
C	.328	.308	8.33	7.82	M	.66	---	16.8	---
D	.646	.626	16.4	15.9	N	.130 dia	.110 dia	3.30 dia	2.79 dia
E	.646	.626	16.4	15.9	P	.21	.15	5.3	3.8
F	1.30	1.24	33.0	31.5	Q	.32	.26	8.1	6.6
G	.35	---	8.9	---	R	.340	.320	8.64	8.13
H	.28	.22	7.1	5.6	S	.69	.63	17.5	16.0
J	.59	.53	15.0	13.5					

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.
4. Round corners of case may be squared.

FIGURE 3. PIN M3928/20-04.

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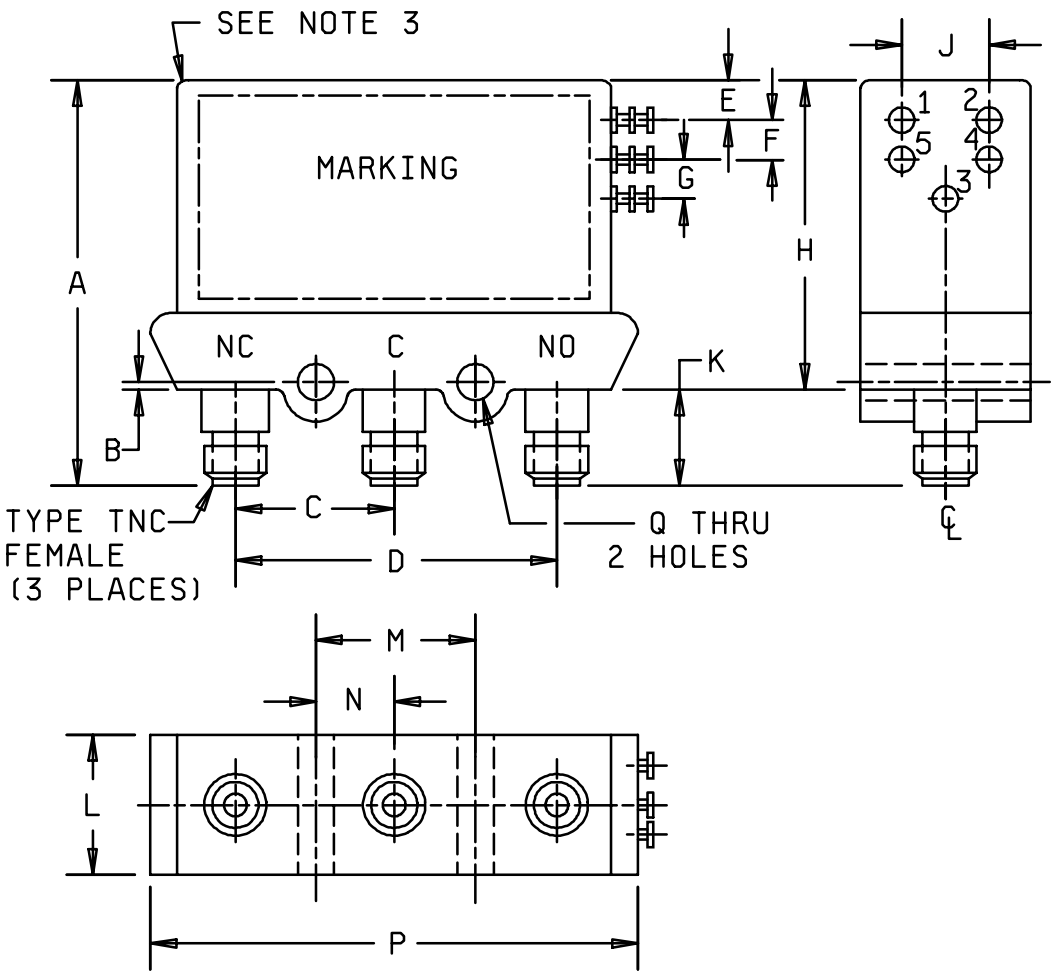
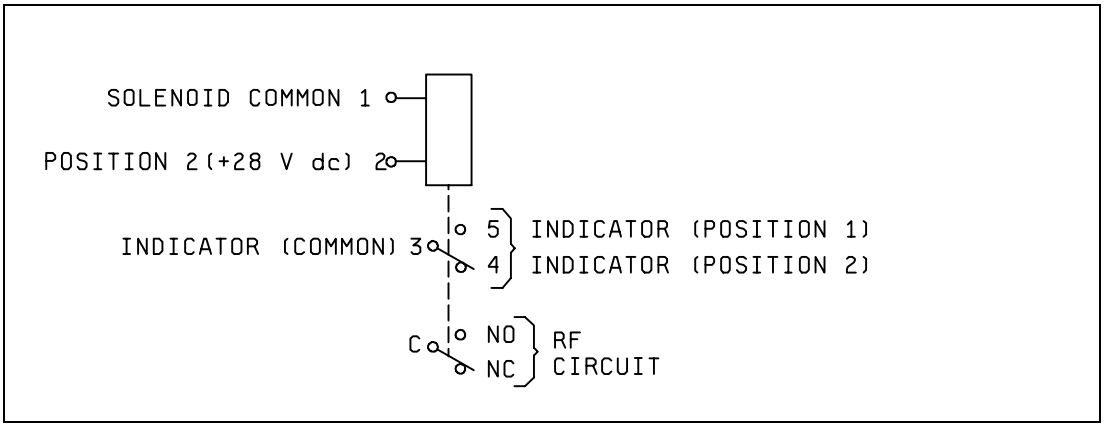


FIGURE 4. PIN M3928/20-05.

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Position 1. (Deenergized or failsafe position). C connects to NC connector.
NO is open circuit.
Position 2. (Energized position). C connects to NO connector. NC is open circuit.

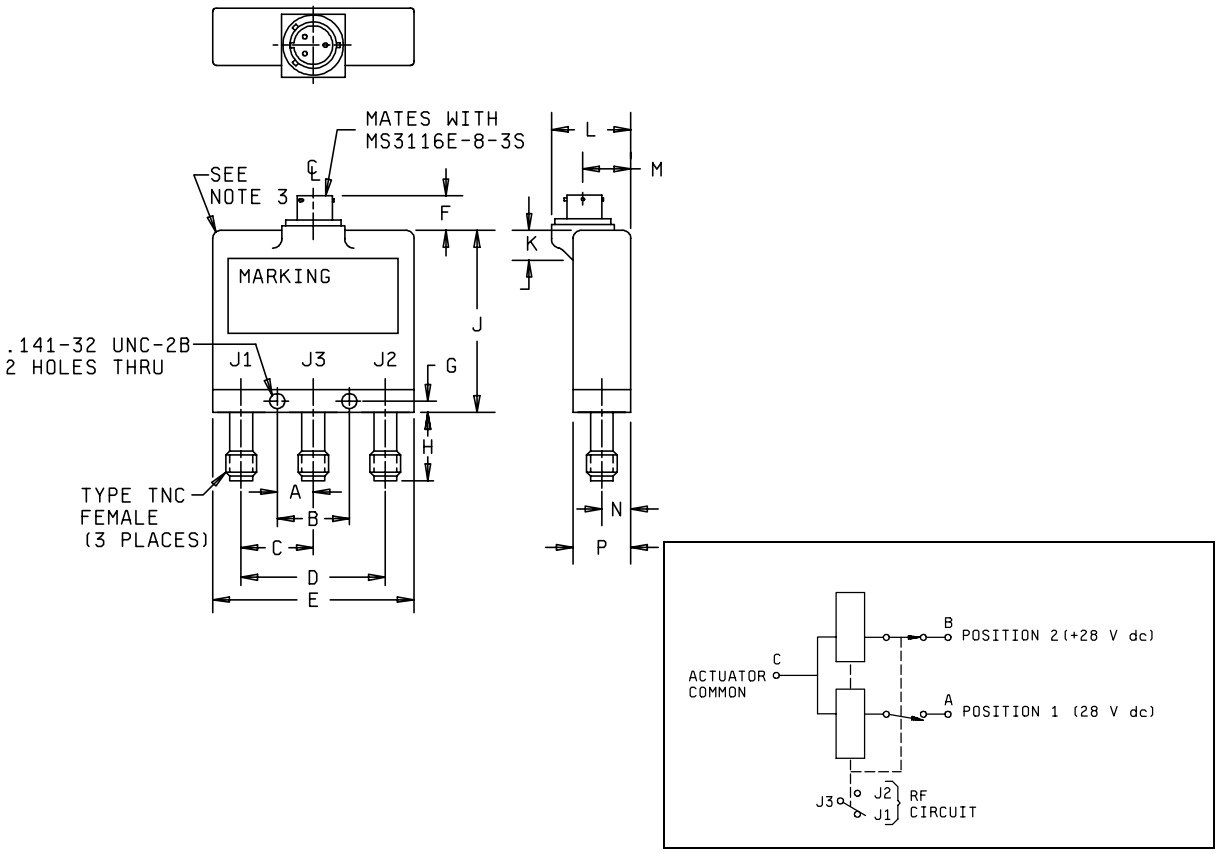
SCHEMATIC (SHOWN IN POSITION 1)

Dimensions									
Letter	Inches		Millimeters		Letter	Inches		Millimeters	
	Max	Min	Max	Min		Max	Min	Max	Min
A	2.23	2.17	56.6	55.1	H	1.70	---	43.2	---
B	.013	.007	0.33	0.18	J	.310	.290	7.87	7.37
C	.890	.870	22.61	22.10	K	.50	---	12.7	---
D	1.770	1.750	44.96	44.45	L	.90	---	22.9	---
E	.310	.290	7.87	7.37	M	.910	.890	23.11	22.61
F	.260	.240	6.60	6.10	N	.460	.440	11.68	11.18
G	.260	.240	6.60	6.10	P	2.75	---	69.8	---
					Q	.151 dia	.131 dia	3.84 dia	3.33 dia

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.
4. Round corners of case may be squared.

FIGURE 4. PIN M3928/20-05 - Continued.

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Check the source to verify that this is the current version before use.



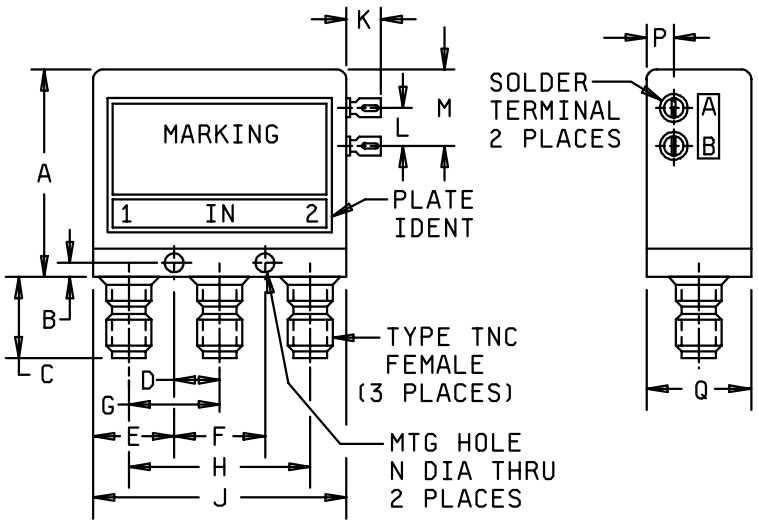
Position 1. Energize pin A. J3 connects to J1, J2 is open circuit.
Position 2. Energize pin B. J3 connects to J2, J1 is open circuit.
SCHEMATIC (SHOWN IN POSITION 1)

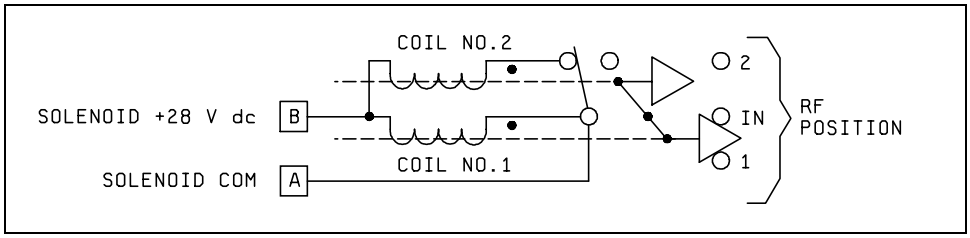
Dimensions									
Letter	Inches		Millimeters		Letter	Inches		Millimeters	
	Max	Min	Max	Min		Max	Min	Max	Min
A	.328	.308	8.33	7.82	H	.66	---	16.8	---
B	.646	.626	16.41	15.90	J	1.80	---	45.7	---
C	.646	.626	16.41	15.90	K	.30	---	7.6	---
D	1.30	1.24	33.0	31.5	L	.78	.72	19.8	18.3
E	1.83	1.77	46.5	45.0	M	.43	.37	10.9	9.4
F	.55	.49	14.0	12.4	N	.340	.320	8.64	8.13
G	.12	.06	3.0	1.5	P	.69	.63	17.5	16.0

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.
4. Round corners of case may be squared.

FIGURE 5. PIN M3928/20-06.

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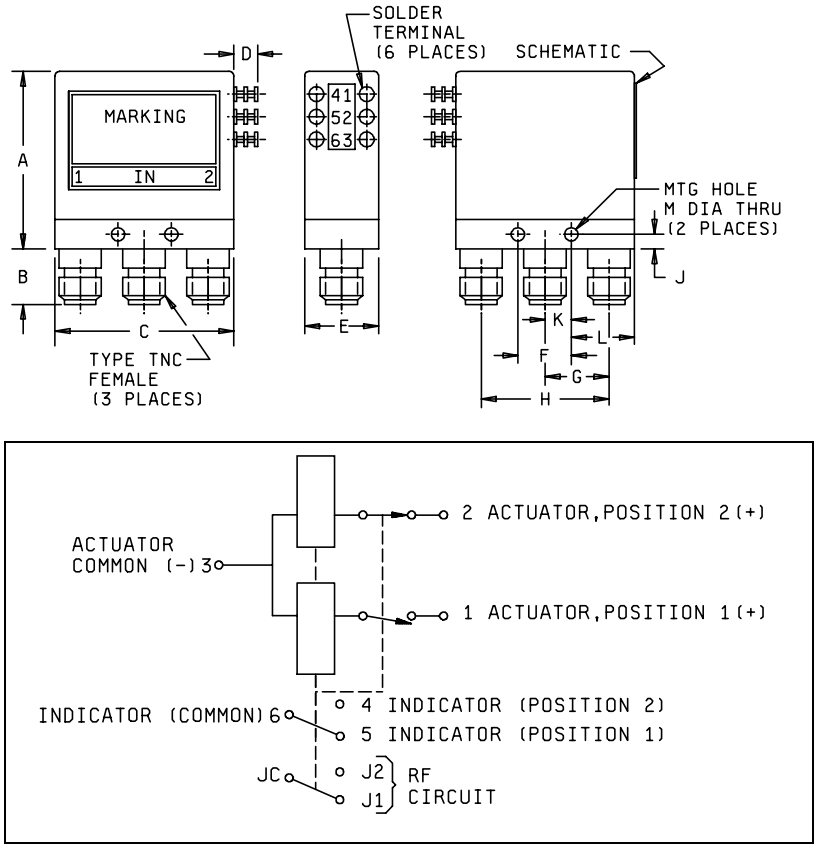
Shown in failsafe position (position 1)
Energized term B to connect in to 2 (position 2).

Dimensions									
Letter	Inches		Millimeters		Letter	Inches		Millimeters	
	Max	Min	Max	Min		Max	Min	Max	Min
A	2.07	---	52.6	---	J	2.65	---	67.3	---
B	.210	.190	5.33	4.83	K	.32	---	8.1	---
C	.59	.53	15.0	13.5	L	.38	.32	9.7	8.1
D	.452	.432	11.48	10.97	M	.73	.67	18.5	17.0
E	.880	.860	22.35	21.84	N	.213 dia	.193 dia	5.41 dia	4.90 dia
F	.895	.875	22.73	22.23	P	.27	.21	6.9	5.3
G	.895	.875	22.73	22.23	Q	1.09	---	27.7	---
H	1.780	1.760	45.21	44.70					

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.
4. Round corners of case may be squared.

FIGURE 6. PIN M3928/20-07.

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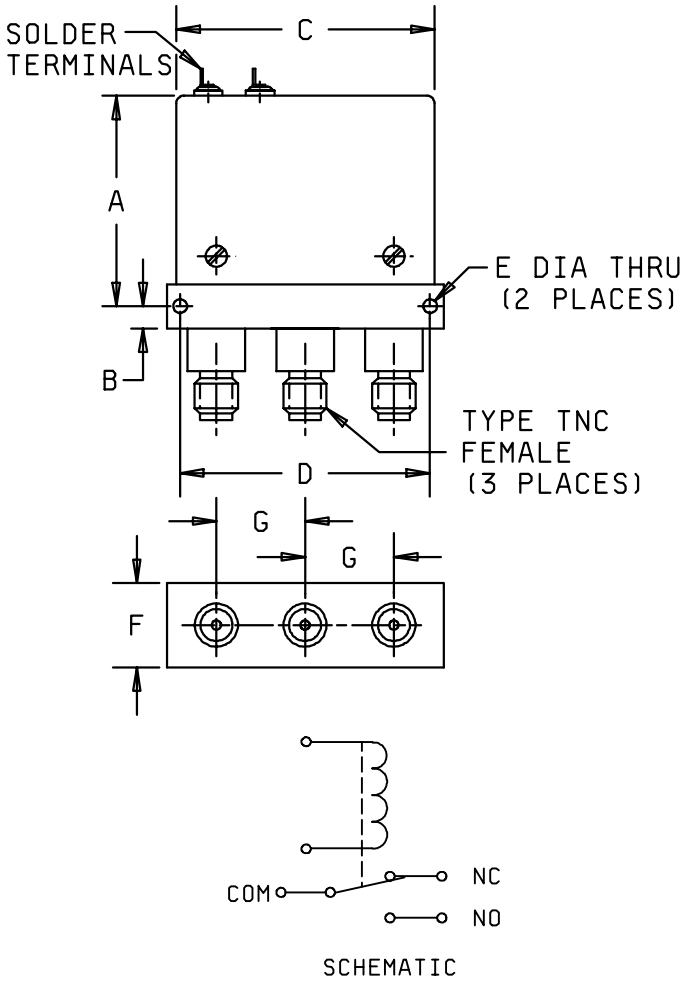
Position 1. Energize terminal 1. JC connects to J1, J2 is open circuit.
Position 2. Energize terminal 2. JC connects to J2, J1 is open circuit.
Schematic (shown in position 1)

Dimensions									
Letter	Inches		Millimeters		Letter	Inches		Millimeters	
	Max	Min	Max	Min		Max	Min	Max	Min
A	2.07	---	52.6	---	G	.895	.875	22.73	22.23
B	.59	.53	15.0	13.5	H	1.780	1.760	45.21	44.70
C	2.65	---	67.3	---	J	.210	.190	5.33	4.83
D	.32	---	8.1	---	K	.452	.432	11.48	10.97
E	1.09	---	27.7	---	L	.880	.860	22.35	21.84
F	.895	.875	22.73	22.23	M	.213 dia	.193 dia	5.41 dia	4.90 dia

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.
4. Round corners of case may be squared.

FIGURE 7. PIN M3928/20-08.

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SCHEMATIC

Dimensions									
Letter	Inches		Millimeters		Letter	Inches		Millimeters	
	Max	Min	Max	Min		Max	Min	Max	Min
A	2.135	2.115	54.23	53.72	E	.190 dia	.170 dia	4.83 dia	4.32 dia
B	.369	.349	9.37	8.86	F	1.03	.97	26.2	24.6
C	2.698	2.678	68.53	68.02	G	.885	.865	22.48	21.97
D	2.385	2.365	60.58	60.07					

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 inch = 25.4 mm).
3. Unless otherwise specified, tolerances are ± .010 (± 0.25 mm) for three place decimals and ± .03 (±0.8 mm) for two place decimals.
4. Round corners of case may be squared.

FIGURE 8. Switch configuration and schematic, PIN M3928/20-09.

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TABLE I. Electrical and performance characteristics. 1/

PIN M3928/20-	Fig. No.	Housing	Frequency range dc to GHz	VSWR	Insertion loss (dB)	Isolation (dB)	Switch time (ms)	Position indication circuit and rating	Life cycles x 1000	Fail-safe or latching	Operating current (A) 2/	Holding current (A) 2/	Nominal operating voltage	Pickup voltage	Dropout voltage	Power and indicator connector	Weight (oz)
				Max	Max	Min	Max				Max	Max					Max
02N, S	2	E	6	15:1	0.3	50	25	None	1000	F	0.12	0.12	28 V dc	20 V dc	5 V dc	Solder terminals	4
03N, S	3	E	11	1.4:1	0.4	60	20	None	100	F	0.8	0.4	28 V dc	18 V dc	5 V dc	MS3113-8-2P	10
04N, S 4/	4	E	11.25	1.4:1	0.5	50	20	0.2A at 40 V dc	100	F	0.28	0.28	28 V dc	22 V dc	5 V dc	Solder terminals	4
05N, S	5	E	0.4	1.2:1	0.15	60	25	2A at 28 V dc	1000	F	0.25	0.25	28 V dc	20 V dc	5 V dc	Solder terminals	6
06N, S 3/ 4/	6	E	12.4	1.5:1	0.5	50	20	None	1000	L	0.56	0	28 V dc	24 V dc	---	Mates with MS3116E-8-3S	4
07N, S	7	E	12.4	1.5:1	0.5	60	20	None	100	F	0.80	0.40	28 V dc	20 V dc	10 V dc	Solder terminals	9.5
08N, S 3/ 5/	8	E	12.4 at 1.57	1.5:1 1.2:1	0.5 0.15	50 60	20 30	0.1A at 28 V dc	100	L	0.56	---	28 V dc	20 V dc	---	Solder terminals	4
09N, S	9	E	12.4	1.5:1	.50	60	15	None	1000	F	0.12	0.12	28 V dc	20 V dc	8 V dc	Solder terminals	7

1/ M3928/20-01 has been deleted.
2/ At 28 V dc and 20 deg C.
3/ Transient interference (RFI).
4/ Sequence: Make-before-break.
5/ RF power handling capacity: 80 watts minimum at sea level and 10^{-5} through 10^{-10} torr.

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REQUIREMENTS:

Dimensions and configurations: See figures 1 through 8.

RF connectors: Female connectors (3 places) shall meet the requirements of MIL-PRF-39012 and shall mate with TNC type male connectors in accordance with MIL-PRF-39012/26.

Electrical and performance characteristics: See table I.

RF power handling capability (average): 50 watts (minimum).

Nominal impedance: 50 ohms.

Termination: Open.

Vibration: Method I.

Operating temperature: -55 deg C to +85 deg C.

Part or Identifying Number (PIN): M3928/20- (and dash number from table I).

Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5985-1209-11)

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